Follow-up Survey 2023 Documentation

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The winter of 2022/23 saw a large increase in energy prices in the UK. The SERL team developed a survey to assess energy use practices during this winter. The design and fieldwork of the survey was funded by the CREDS UKRI grant (EP/R035288/1), the Project VENICE Network Innovation Allowance (NIA) grant (NIA_WPD_059) and the SERL UKRI grant (EP/P032781/1).

The survey was sent in the post to 12,001 households on 2 February 2023. A weblink was included to give householders the opportunity to complete the survey online. A reminder was sent on 24 February 2023 with no additional paper copy included. Survey distribution and data entry were handled by a survey company. There was no payment associated with the survey. About 5,400 responses were received.

An overview of the findings from the survey can be found in *Self-reported energy use behaviour changed significantly during the cost-of-living crisis in winter 2022/23: insights from cross-sectional and longitudinal surveys in Great Britain*.

The postal data entry was a direct copy of the online survey so all logic was applied consistently on both. This means that if a response was provided on the paper version to a question which the online branching from the previous question would not have allowed, then this answer was not entered. If surveys were returned with missing pages, the relevant questions were left blank. If the respondent was prompted to provide one response but actually marked multiple responses, no response was recorded. A difference in the data compared to previous surveys is that nearly all answers are text rather than numeric format with the full survey answer provided e.g. “Always work from home”.

Alongside the survey data provided with this SERL Observatory data edition, we also provide a copy of the postal survey. The relevant file is:

- *serl_follow_up_survey_2023_data_edition{number}.csv*, where *{number}* is the number of the edition, *e.g.* "06"

The fields in the survey are listed in the table below. The logic for replacing blanks in the primary data and for derived error flags is described in the relevant notes.

Survey field	Approximate question	Possible values	Variable type	Notes
PUPRN	Pseudonymised participant identifier		Assigned	
A1	*During this winter, to what temperature do you set your heating controller for late afternoons or evenings?*			
A1_units	Units corresponding to numeric answer	Celsius; Fahrenheit; Don't know/can't do this	Primary	Note 1
A1_degC	Numeric answer in Celsius box	Numeric	Primary	Note 2
A1_degF	Numeric answer in Fahrenheit box	Numeric	Primary	Note 2
A1_corr_C	Converted / corrected Celsius temperature	Numeric	Primary	Note 2
A1_edit		TRUE/FALSE	Derived	Note 2
A1_err		TRUE/FALSE	Derived	Note 2
A2	During this winter, are there any living spaces (e.g., bedrooms, living / dining rooms, kitchens) in your accommodation that your household does not normally heat?	Yes; No; No response	Primary	Note 1
A3	During this winter, have you reduced the flow temperature of your boiler?	Yes; No; No response	Primary	Note 1
A4	During this winter, are you heating your house for fewer hours than in previous winters?	Yes; No; No response	Primary	Note 1
A5	*This winter, how often do you personally do each of the following?*			
A5_1	Switch off lights in rooms that aren't being used	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1

A5_2	Put more clothes on when feeling cold rather than putting the heating on or turning it up	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_3	Turn down your thermostat or turn off your heating when you leave the house	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_4	Use a standalone heater rather than putting the heating on or turning it up	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_5	Use an electric blanket, hot water bottle or similar when feeling cold rather than putting the heating on or turning it up	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_6	Turn down radiators when rooms are not in use	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_7	Turn down radiators in rooms that are in use	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_8	Run the washing machine with a full load rather than running it part full	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_9	Run the washing machine at 30 degrees or lower	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_10	Dry clothes without using a tumble dryer	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_11	Turn appliances off standby when they are not in use	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_12	Close curtains / blinds at night	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_13	Take a shower rather than having a bath	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_14	Take short showers rather than longer showers	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_15	Use the dishwasher rather than washing up	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A5_16	Avoid using the cooker or oven when preparing a main meal	Always; Very often; Quite often; Not very often; Never; Not applicable, cannot do this	Primary	Note 1
A6	*To what extent have you done the following this winter compared with last winter?*			

A6_1	Switched off lights in rooms that aren't being used	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_2	Put more clothes on when feeling cold rather than putting the heating on or turning it up	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_3	Turned down your thermostat or turn off your heating when you leave the house	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_4	Used a standalone heater rather than putting the heating on or turning it up	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_5	Used an electric blanket, hot water bottle or similar when feeling cold rather than putting the heating on or turning it up	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_6	Turned down radiators when rooms are not in use	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_7	Turned down radiators in rooms that are in use	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_8	Ran the washing machine with a full load rather than running it part full	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_9	Ran the washing machine at 30 degrees or lower	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_10	Dried clothes without using a tumble dryer	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_11	Turned appliances off standby when they are not in use	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_12	Closed curtains / blinds at night	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1

A6_13	Taken a shower rather than having a bath	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_14	Taken short showers rather than longer showers	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_15	Used the dishwasher rather than washing up	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A6_16	Avoided using the cooker or oven when preparing a main meal	A lot more; A little more; About the same; A little less; A lot less; Not applicable, cannot do this	Primary	Note 1
A7	When your accommodation is unoccupied for more than a day or so, how often will your household adjust the heating controls to ensure the heating either won't, or is much less likely to come on?	Always; Very often; Quite often; Not very often; Never; Not applicable	Primary	Note 1
A8	Other than central heating, does your household use any standalone heaters in your accommodation?	Yes; No; No response	Primary	Note 1
A9	Do any of your standalone heaters have their own source of fuel?	Yes, some or all have their own source of fuel (e.g., logs, coal, bottled gas etc.); No, they are all powered by mains gas or electricity; No response; NA	Primary	Note 4
A10	In your opinion, during very cold winter weather, how often are these standalone heaters typically used in your household?	Daily; Most days; Rarely - only if I/we really have to; Never; Varies - depends on temperature or other reasons; Don't know	Primary	Note 4
A11	Compared to before this winter, are you using your smart meter In-Home Display/Smart Energy Display?	More often; Less often; About the same; I don't have this; It is not working; Don't know	Primary	Note 1
A12	How much effort, if any, would you say your household makes	A great deal of effort; Some effort; A little effort; No effort at all; Don't know	Primary	Note 1

	to limit or reduce the amount of gas or electricity used?			
B1	How many bathrooms do you have with a plumbed in bath or shower?	0; 1; 2; 3; 4 or more	Primary	Note 1
B2	*What type of central heating does your accommodation have?*			
B2_1_yes	No central heating	Yes; No; No response	Primary	Note 3
B2_2_yes	Gas e.g., gas boiler	Yes; No; No response	Primary	Note 3
B2_3_yes	Electric storage heaters (not electric radiators)	Yes; No; No response	Primary	Note 3
B2_4_yes	Electric radiators (not electric storage heaters)	Yes; No; No response	Primary	Note 3
B2_5_yes	Heat pump	Yes; No; No response	Primary	Note 3
B2_6_yes	Other electric	Yes; No; No response	Primary	Note 3
B2_7_yes	Oil	Yes; No; No response	Primary	Note 3
B2_8_yes	Solid fuel, e.g. wood or coal	Yes; No; No response	Primary	Note 3
B2_9_yes	Biomass for boiler	Yes; No; No response	Primary	Note 3
B2_10_yes	District or community heating	Yes; No; No response	Primary	Note 3
B2_11_yes	Other	Yes; No; No response	Primary	Note 3
B2_1__add_rep	No central heating	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_2__add_rep	Gas e.g., gas boiler	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_3__add_rep	Electric storage heaters (not electric radiators)	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_4__add_rep	Electric radiators (not electric storage heaters)	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_5__add_rep	Heat pump	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_6__add_rep	Other electric	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_7__add_rep	Oil	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3

B2_8__add_rep	Solid fuel, e.g. wood or coal	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_9__add_rep	Biomass for boiler	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_10__add_rep	District or community heating	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_11__add_rep	Other	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B2_1_err	No central heating	TRUE/FALSE	Derived	Note 3
B2_2_err	Gas e.g., gas boiler	TRUE/FALSE	Derived	Note 3
B2_3_err	Electric storage heaters (not electric radiators)	TRUE/FALSE	Derived	Note 3
B2_4_err	Electric radiators (not electric storage heaters)	TRUE/FALSE	Derived	Note 3
B2_5_err	Heat pump	TRUE/FALSE	Derived	Note 3
B2_6_err	Other electric	TRUE/FALSE	Derived	Note 3
B2_7_err	Oil	TRUE/FALSE	Derived	Note 3
B2_8_err	Solid fuel, e.g. wood or coal	TRUE/FALSE	Derived	Note 3
B2_9_err	Biomass for boiler	TRUE/FALSE	Derived	Note 3
B2_10_err	District or community heating	TRUE/FALSE	Derived	Note 3
B2_11_err	Other	TRUE/FALSE	Derived	Note 3
B3	*Which of the following does your accommodation have?*			
B3_1_yes	Solar panels for electricity	Yes; No; No response	Primary	Note 3
B3_2_yes	Solar water heating	Yes; No; No response	Primary	Note 3
B3_3_yes	Battery storage that is wired into your mains electricity	Yes; No; No response	Primary	Note 3
B3_4_yes	Electric vehicle (EV) charging point	Yes; No; No response	Primary	Note 3
B3_5_yes	LED light bulbs in most of your lights	Yes; No; No response	Primary	Note 3
B3_6_yes	Radiators in main living areas	Yes; No; No response	Primary	Note 3
B3_7_yes	Thermostatic radiator valves on at least half of your radiators	Yes; No; No response	Primary	Note 3
B3_8_yes	Underfloor heating in main living areas	Yes; No; No response	Primary	Note 3

B3_9_yes	Warm air heating units in main living areas	Yes; No; No response	Primary	Note 3
B3_10_yes	Central heating thermostat	Yes; No; No response	Primary	Note 3
B3_11_yes	Central heating time clock / programmer	Yes; No; No response	Primary	Note 3
B3_12_yes	Smart heating control (e.g., Nest, Hive, Tado)	Yes; No; No response	Primary	Note 3
B3_13_yes	Hot water cylinder	Yes; No; No response	Primary	Note 3
B3_14_yes	Insulating jacket on hot water cylinder	Yes; No; No response	Primary	Note 3
B3_15_yes	Thermostat on hot water cylinder	Yes; No; No response	Primary	Note 3
B3_16_yes	Other	Yes; No; No response	Primary	Note 3
B3_1_add_rep	Solar panels for electricity	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_2_add_rep	Solar water heating	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_3_add_rep	Battery storage that is wired into your mains electricity	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_4_add_rep	Electric vehicle (EV) charging point	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_5_add_rep	LED light bulbs in most of your lights	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_6_add_rep	Radiators in main living areas	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_7_add_rep	Thermostatic radiator valves on at least half of your radiators	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_8_add_rep	Underfloor heating in main living areas	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_9_add_rep	Warm air heating units in main living areas	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_10_add_rep	Central heating thermostat	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_11_add_rep	Central heating time clock / programmer	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_12_add_rep	Smart heating control (e.g., Nest, Hive, Tado)	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3

B3_13_add_rep	Hot water cylinder	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_14_add_rep	Insulating jacket on hot water cylinder	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_15_add_rep	Thermostat on hot water cylinder	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_16_add_rep	Other	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B3_1_err	Solar panels for electricity	TRUE/FALSE	Derived	Note 3
B3_2_err	Solar water heating	TRUE/FALSE	Derived	Note 3
B3_3_err	Battery storage that is wired into your mains electricity	TRUE/FALSE	Derived	Note 3
B3_4_err	Electric vehicle (EV) charging point	TRUE/FALSE	Derived	Note 3
B3_5_err	LED light bulbs in most of your lights	TRUE/FALSE	Derived	Note 3
B3_6_err	Radiators in main living areas	TRUE/FALSE	Derived	Note 3
B3_7_err	Thermostatic radiator valves on at least half of your radiators	TRUE/FALSE	Derived	Note 3
B3_8_err	Underfloor heating in main living areas	TRUE/FALSE	Derived	Note 3
B3_9_err	Warm air heating units in main living areas	TRUE/FALSE	Derived	Note 3
B3_10_err	Central heating thermostat	TRUE/FALSE	Derived	Note 3
B3_11_err	Central heating time clock / programmer	TRUE/FALSE	Derived	Note 3
B3_12_err	Smart heating control (e.g., Nest, Hive, Tado)	TRUE/FALSE	Derived	Note 3
B3_13_err	Hot water cylinder	TRUE/FALSE	Derived	Note 3
B3_14_err	Insulating jacket on hot water cylinder	TRUE/FALSE	Derived	Note 3
B3_15_err	Thermostat on hot water cylinder	TRUE/FALSE	Derived	Note 3
B3_16_err	Other	TRUE/FALSE	Derived	Note 3
B4	*Which of the following kinds of insulation does your accommodation have?*			
B4_1_yes	Loft insulation	Yes; No; No response	Primary	Note 3

B4_2_yes	Cavity wall insulation	Yes; No; No response	Primary	Note 3
B4_3_yes	Solid wall insulation	Yes; No; No response	Primary	Note 3
B4_4_yes	Underfloor insulation	Yes; No; No response	Primary	Note 3
B4_5_yes	Double or secondary glazed windows	Yes; No; No response	Primary	Note 3
B4_6_yes	Draughtproofing on windows, doors and/or chimneys	Yes; No; No response	Primary	Note 3
B4_7_yes	Other	Yes; No; No response	Primary	Note 3
B4_1_add_rep	Loft insulation	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_2_add_rep	Cavity wall insulation	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_3_add_rep	Solid wall insulation	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_4_add_rep	Underfloor insulation	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_5_add_rep	Double or secondary glazed windows	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_6_add_rep	Draughtproofing on windows, doors and/or chimneys	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_7_add_rep	Other	Has been added or replaced in the last 12 months; No; NA	Primary	Note 3
B4_1_err	Loft insulation	TRUE/FALSE	Derived	Note 3
B4_2_err	Cavity wall insulation	TRUE/FALSE	Derived	Note 3
B4_3_err	Solid wall insulation	TRUE/FALSE	Derived	Note 3
B4_4_err	Underfloor insulation	TRUE/FALSE	Derived	Note 3
B4_5_err	Double or secondary glazed windows	TRUE/FALSE	Derived	Note 3
B4_6_err	Draughtproofing on windows, doors and/or chimneys	TRUE/FALSE	Derived	Note 3
B4_7_err	Other	TRUE/FALSE	Derived	Note 3
B5	Do you have any problems with condensation, damp or mould in your home?	Yes; No; Don't know	Primary	Note 5
B6	*Do you have any of the following?*			

B6_1	Mould around the windows or in the bathroom	Minor; Substantial; Don't know	Primary	Note 5
B6_2	Mould on any walls	Minor; Substantial; Don't know	Primary	Note 5
B6_3	Mould on furnishings	Minor; Substantial; Don't know	Primary	Note 5
C1	Select option that corresponds with your recent gross household annual income?	Below £10,000; £10,001 to £20,000; £20,001 to £30,000; £30,001 to £40,000; £40,001 to £50,000; £50,001 to £60,000; £60,001 to £70,000; £70,001 to £80,000; £80,001 to £90,000; £90,0001 to £100,000; Above £100,000; Prefer not to answer	Primary	Note 1
C2	*Which of these methods do you use to pay for your energy?*			
C2_electricity	Electricity	Direct debit (including online direct debit); Payment on receipt of bill (by post, telephone, online or at bank/post office); Pre-payment meter; Included in rent; Other; Not applicable / no mains gas; Don't know	Primary	Note 1
C2_gas	Gas	Direct debit (including online direct debit); Payment on receipt of bill (by post, telephone, online or at bank/post office); Pre-payment meter; Included in rent; Other; Not applicable / no mains gas; Don't know	Primary	Note 1
C3	During the cold winter weather, can you normally keep comfortably warm in your living room?		Primary	Note 1
C4	*Did you answer 'No' to the previous question for any of the following reasons?*			
C4_1	You feel your home is difficult to heat	You feel your home is difficult to heat: No response; NA	Primary	Note 6
C4_2	You feel it is difficult to afford the fuel to heat your home	You feel it is difficult to afford the fuel to heat your home; No response; NA	Primary	Note 6
C4_3	Prefer not to say	Prefer not to say; No response; NA	Primary	Note 6
C4_4	None of the above	None of the above; No response; NA	Primary	Note 6
C4_5	Other reason	Other reason; No response; NA	Primary	Note 6
C5	How easy or difficult is it for you to meet your heating/fuel costs?	Very easy; Fairly easy; Neither easy nor difficult; Fairly difficult; Very difficult; Don't know	Primary	Note 1

C6	During this winter, do you leave your home to go to heated spaces elsewhere with the main purpose of keeping warm?	Daily; Most days; Rarely; never; Don't know; Prefer not to say	Primary	Note 1
D1	How many people currently live in your household, including you?		Primary	Note 7
D1_flag		TRUE/FALSE	Derived	Note 7
D2	*Including you, how many people are there in each of the following age groups in your household?*			
D2_0_5	0-5 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_6-15	6-15 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_16-24	16-24 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_24_44	25-44 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_45-64	45-64 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_65_74	65-74 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_75_85	75-84 years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_85plus	85+ years	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 7
D2_ignored		TRUE/FALSE	Derived	Note 7
D2_min_total			Derived	Note 7
D3	*Thinking about the working situation of each member of your household aged 16 and over, including you, how many people would you say fall into each category below?*			
D3_1	Working (paid or unpaid): 30 hours a week or more	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 8
D3_2	Working (paid or unpaid): less than 30 hours a week	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 8
D3_3	Not working because of long term sickness or disability	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 8

D3_4	Unemployed but seeking work	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 8
D3_5	Student	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 8
D3_6	Retired/at home/not seeking work (including looking after the home or family)	0 people; 1 person; 2 people; 3 people; 4 or more people	Primary	Note 8
D3_7	Other		Primary	Note 8
D3_flag_high		TRUE/FALSE	Derived	Note 8
D3_flag_low		TRUE/FALSE	Derived	Note 8
D3_ignored		TRUE/FALSE	Derived	Note 8
D3_total_over16			Derived	Note 8
D4	Does anyone in your household work in the following ways?	Always work from home; Sometimes work from home; Never work from home; Not applicable /prefer not to say	Primary	Note 9
D4_flag		TRUE/FALSE	Derived	Note 9
D5	How many plug-in electric vehicles does your household have?	0; 1; 2; 3 or more; Don't know	Primary	Note 1
D6	How many of these vehicles has your household acquired in the last 12 months?	0; 1; 2; 3 or more; Don't know	Primary	Note 1
D6_err		TRUE/FALSE	Derived	Note 10
E1	How well would you say you are managing financially these days?	Living comfortably; Doing alright; Just about getting by; Finding it quite difficult; Finding it very difficult; Don’t know; Prefer not to say	Primary	Note 1
E2	Overall, how satisfied are you with your life nowadays, where 0 is 'not at all satisfied' and 10 is 'completely satisfied'?	0;1;2;3;4;5;6;7;8;9;10	Primary	Note 1
E3	Overall, to what extent do you feel that the things you do in your life are worthwhile, where 0 is 'not at all worthwhile' and 10 is 'completely worthwhile'?	0;1;2;3;4;5;6;7;8;9;11	Primary	Note 1
Collection method		Online; Postal	Primary	

Note 1
Where there is 'no tick in a box' for a question in the survey, the blank "" answer is changed to "No response" for all questions where an answer is expected but has not been given.

Note 2 Question A1: temperature

Any decimals in postal survey were rounded to nearest whole number (19.5 becomes 20, 19.1 becomes 19), only whole numbers could be entered online.
Column A1_corr_C updates temperature in °C, A1_edit is flagged TRUE for each case where this corrected value is not equal either to A1_deg_C or to A1_deg_F converted to °F.

Case	**Action**
A1_degC in expected band (10°C to 35°C)	A1_corr_C = A1_deg_C
A1_deg_C below lower Celsius limit	A1_err set to TRUE
A1_deg_C above upper Celsius limit and below upper Fahrenheit limit: assumed to be °F	A1_corr_C is value converted from F.
A1_deg_C above upper Celsius limit but below lower Fahrenheit limit (i.e. between 35 and 50)	No corrections made. A1_err set to TRUE
A1_deg_C above upper Fahrenheit limit	A1_err set to TRUE.
A1_degF within expected Fahrenheit band (50°F to 95°F),	A1_deg_C = A1_deg_F converted to Fahrenheit
A1_deg_F below lower Fahrenheit limit and above lower Celsius limit: assumed to be Celsius	A1_corr_C = A1_deg_F.
A1_deg_F below lower Fahrenheit limit and below lower Celsius limit	A1_err set to TRUE
A1_deg_F above upper Fahrenheit limit	A1_err set to TRUE

Where a range is apparently entered (e.g.A1_degC= 2022) a midpoint is chosen (in this hypothetical case A1_corr_C= 21°C).
Where a decimal point seems to be missing this is corrected (e.g. A1_degC=205: A1_corr_C =20.5).

Note 3 Question B2/B3/B4: questions with "added or replaced in last 12 months" options
These are "please tick all that apply" questions so more than one answer is possible. The total number of heating systems selected is shown in column B2_total.

The table below shows the logic for all questions where there is an added/replaced in last 12 months option, B2_2 quoted as an example.

LH column (e.g. B2_2_yes)	**RH column (e.g. B2_2_add_rep)**	**Action**
"Yes"	"Yes"	No change
"Yes"	""	RH column set to "NA"
""	"Yes"	LH column set to "No response". error flag (e.g. B2_2_err) set to TRUE
""	""	LH column set to "No" RH column set to "No"

Note 4 Logic for blank answers for 'branching' questions A9/A10

A8	**A9**	**A10**	**Action**
""	""	""	A9 coded "No response" A10 coded "No response"
No	""	""	A9/A10 coded NA
Yes	""	""	A9 coded "No response" A10 coded "No response"
Yes	"powered by mains gas or electricity"	""	A10 coded NA
Yes	"some or all have own source of fuel"	""	A10 coded "No response"
Yes	"some or all have own source of fuel"	One of 6 answers about frequency of use	

Note 5 Logic for 'branching' questions B5/B6

B5	B6_1 / B6_2 /B6_3	
""	""	
No	""	B6_1 etc coded NA
Don't know	""	B6_1 etc coded NA
Yes	All ""	B6_1 etc coded "No response"
Yes	Any of possible answers to B6_1 etc marked	All other answers to B6_1 etc coded NA "No".

Some respondents ticked Don't know for B5 and then selected one or more Don't know options for B6 – this means that the number of NA answers varies across B6_1 to B6_3.

Note 6 Logic for 'branching' questions C3/C4_N

C3	C4_1, C4_2, C4_3, C4_4, C4_5	
""	""	C4_N coded NA
Yes	""	C4_N coded NA
Don't know	""	C4_N coded NA
No	""	C4_N coded "No response"
No	One of 5 answers	All C4_N which are blank coded "No"

Note 7 questions D1 and D2

The answer to D1 is numeric, so no response is left as NA rather than set to "No response" which would change the data type of this column.

D2_min_total - minimum number of people - is calculated from sum of answers to D2 (if any "4 or more people" options are selected, exact total cannot be determined).

D1_flag is set to TRUE if:

- D1 is NA (i.e. no number entered)
- D1 does not match D2_min_total (If answer for one or more age categories in D2 is "has_4_or more" and D2_min_total <D1, D1 could still be correct, however this applies to an extremely small number of cases and extra logic to investigate this has not been included). In cases where D1=0 or "", D2_min_total could be used as an alternative indicator of household size.

If at least one of the D2 answers is ticked, all blanks coded to "0 People".

If no boxes at all are ticked then all answers set to "No response".

Note 8 question D3

If at least one of the D3 answers is ticked, all blanks coded to "0 People".

If no boxes at all are ticked then all answers set to "No response".

D3_flag_high is set to TRUE if the total of those over 16 in the household calculated from D2 is less than the total number of adults D3_total_over16 (this may be a result of 4 or more in an age group).
D3_flag_high is set to TRUE if the total of those over 16 in the household calculated from D2 is greater than the total number of adults D3_total_over16.

Note 9 Question D4
There is scope for confusion on this question, if several people in the household work but with different patterns. "" is coded as "no response" but may be a result of ticking more than one box in postal survey (frequency of "no response" roughly 4 times higher in postal than online answers).
D4_flag is TRUE if no-one is working from D3 (sum of D3_1, D3_2 and D3_5) but there is a response to D4.

Note 10 Question D6
D6_err is set to TRUE if the numeric answer for how many EVs have been acquired in last 12 months is greater than the total number of EVs from D5.